

Speedeat

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A. Introduction

In the beginning, all of us have met some bad experience for eating outside before, so we want to solve the problems about eating outside.

At first, we interview (contextual inquiry) with 7 people and make an affinity diagram. Then, we create three personas to do user analysis. Based on interview, personas and user analysis, we find some common problems for each person. So we the reason behind these problems and describe three main scenarios. According to these problems and scenarios, we envision the function of our system and use storyboard to check the function in those three main scenarios repeatedly. After envisioning, we make a paper prototype to do the user testing. Finally, we also try to implement software prototype on Android system.

B. User Analysis and persona

User Analysis

1. Definition of user group

- Familiar with smartphone and APP
- Usually eat outside (> 7 times per week)
- Busy at weekdays (Don't want to waste time)
- Sensitive about quality of restaurant

2. Checklist of important user characteristics

(1) Physical characteristics

Normal people, but focus on office workers, engineers (ex: 竹科) and the college students.

(2) Knowledge and experience

- Level of education: High school
- Reading level: Well

(3) Computer/Information Technology experience or knowledge

- Computer literacy: Skilled
- Level of experience with similar systems: Smartphone
- Level of experience with other systems: Smartphone, Computer, Laptop

(4) Psychological characteristics

Practical, Open-mind, Impatient, Pay attention to efficiency Curious, Early adopter and Enjoyment

Personas

We defined three personas according to our interview result. The first and second is Taiwanese persona (Victor and Eric). And the third is Foreigner person (James) because the three people of our interviewees are foreigner. The main difference between Taiwanese persona and Foreigner person is language problem. Foreigner person is not good at Chinese, so we need to design bilingual interface (English and Chinese) for our system. The details of each persona are in the following pictures.

1. Primary Persona

Primary Persona



"A man who dares to waste one hour of time has not discovered the value of life."

Demographics

Age: 23
Job: Graduated Student
Status: Single
Location: Hsinchu
Traits: Hard working, Friendly, Smart, Cautious, Thrifty

Personality

Introvert	Extrovert
Thinking	Feeling
Sensing	Intuition
Judging	Perceiving

Motivations

Incentive	High
Fear	Low
Achievement	High
Power	Medium
Social	High

Needs

- Using some app or any method to make the life more efficient.
- Using some app to let him use money wisely.

Hobby

- Play or eat with lots of friends.
- Play basketball and join the team representing for the department.

Bio

My name is Victor. I am a Graduated Student. I really work hard. I not only set the goal to myself but also create a to do list for everyday. And I do my best to reach all of them. I seldom fail to finish it. If I failed, I will reflect and adjust myself. So wasting time is prohibition for me. Beside, I will use some app to keep accounts everyday. Thrifty is also important. Moreover, I really like make lots of friends. I usually play sports, eat, chat and study together which make me life splendid. Sense of accomplishment comes from working and happiness comes from friends.

Goal

- Get the master degree.
- Setting goal must accomplish.
- Making more friends to let me cheerful.
- "To do list" of each day need to be finished.

Frustrations

- Feeling alone.
- Wasting too many time to carry out his work.
- I have lots of meetings with professor and teammate.
- Eating with friends, it is hard to decide what to eat.
- Fail to do something or wasting money because of thoughtless.
- Eating bad or unhealthy food because of hurry.

Pain Points

- Sometimes something is out of his control and make me waste time or money, like traffic or eating.
- Because of many meetings, I don't have much time to enjoy my lunch twice day each week.
- Eating unhealthy or fast food many times because I am hard to know that I don't need to take too much time in which restaurant.

2. Secondary Persona

Secondary Persona



"I think being in love with life is a key to eternal youth."

Demographics

Age: 28
Job: Office worker
Status: Married, no child
Location: Taipei
Traits: Considerate, Funny, Romantic, Curious, Hesitant, Changeable

Personality

Introvert	Extrovert
Thinking	Feeling
Sensing	Intuition
Judging	Perceiving

Motivations

Incentive	Medium
Fear	Medium
Achievement	High
Power	Medium
Social	High

Needs

- Using some method or app to make him work efficiently or prevent from wasting of time.
- The app can recommend him restaurant.

Hobby

- Buy new technical products.
- Like to taste the delicacy and explore new restaurants.
- Usually go out with wife and visit different towns in the weekend.

Bio

Hi, I'm Eric, a software engineer in Neihu Technology Park at Taipei. Because of my background, I like to buy new technical products and enjoy the convenience they bring. I just married with my wife in last year. I really love my wife and want to give her everything I could. We usually went out and visit different towns such as Yilan in the weekend. As well, we love to explore different restaurants, and share them on our instagram. For me, it is important to both enjoy my works and my rest time.

Goal

- Work hard and play hard.
- Enjoy better life with wife.
- Make the balance between work and life.

Frustrations

- Because there is no restaurant at our company, I need to go out for eating with my colleague every lunch. And everyone is hard to decide the restaurant to eat.
- Sometimes, I have the bad experience about food in the unfamiliar restaurant.
- Sometimes, I can't finish my job before evening, so I can't go home and eat dinner with my wife.

Pain Points

- I always waste much time to think about what to eat with co-workers on weekdays and sometime it may take for a long time to wait.
- Some excellent restaurants need to take lots of time for waiting in the weekend.
- Some comment of restaurant on the google may mislead me.

3. Foreigner Persona



C. Task Analysis

Contextual inquiry & affinity diagram

1. Contextual inquiry

(1) Place: Lab, dorm room

(2) People: 7 (4 Taiwanese, 3 Foreigners)

- 1 NTHU EE graduated student, Taiwanese
- 1 NTHU EE undergraduated student, Taiwanese
- 1 Sweden graduated student, Taiwanese
- 1 Computer Engineer, Taiwanese
- 1 NCTU EECS undergraduated student, India
- 1 NCTU Chem PHD student, Polish
- 1 NCTU Postdoc researcher, Polish

(3) Time: 20 minutes for each person

(4) Process:

- Let interviewer imagine that he is going to eat outside right now.
- Let them freely describe their experience of eating outside during working day or holiday.
- We ask them some questions about eating outside which we prepare before interviewing.
- Writing down the results.

(5) Result: <Appendix>

(1) Process:

- (2) Results:

- [illegible]

- Online affinity diagram (paper one revises a little.):



Goals of users

There are two main goal of users to use our system.

- Time: Let users can save more time for eating.
- Quality: Let users can choose the restaurant which they indeed like, even though they never go to eat.

The tasks to be performed

Based on contextual inquiry, persona and user analysis, we found that there are two main problems of our target user. One is time, they may be late, wasting too much time, choose unhealthy food because they only have limited time. The other is quality, they may eat bad food or meet bad service because of they don't have the precise information. So by these problem, we think that the main reason to cause these problems are vacancy, traffic, waiting for food and the comment on the Internet may not really precise. Based on these reasons, we describe three main tasks for our target users.

Task description (Scenarios):

- 1. Someone doesn't want to take too much time to eat.**
- 2. Someone can't late after breakfast, lunch or dinner.**
- 3. Someone in unfamiliar place finds for eating. (ex: Business trip or Travel)**

During these three scenarios, users need to eat as soon as possible and find the good quality restaurant for users.

General description of task characteristics

1. The frequency or timing of the task

(1) Frequency:

Ususally, because most of people need to eat outside everyday and sometimes they may be hurry to do something after eating.

(2) Time constraints: As short as possible.

2. The complexity and difficulty of the task

Everyone usually eat outside everyday. So the tasks are just like daily life. The complexity and difficulty of tasks are not hard. Using our system, user may lower the complexity and difficulty of tasks.

3. The functional requirement of the system

We describe three main scenarios and based on them we envision different function for our system. Actually, we revise these function again and again checking by storytelling.

(1) For scenario 1

- A map with restaurant location
- Connect with Google Maps
- Show the vacancy of restaurant
- Booking service
- Ordering Service
- Showing the order of reserving

(2) For scenario 2

- Same function as previous
- Favorite restaurant list
- Predict the time you need to take in certain restaurant
- The time needed to get there by walking or driving...
- Combine with 2,3,4 to plan a best path and restaurant for you

(3) For scenario 3

- Same function as scenario 2
- Show your friends' comment, score, check in and tag
- Make photos directly from our application and share it on Facebook, IG
- Connect with Google

4. The physical environment of task performance

(1) Where is the task performed? Any place

(2) What other tools does the user have? An app with our system

5. The social, organizational and cultural environment of the task

Our system need to connect with restaurant to update the information of restaurant immediately. The information is like vacancy, number of people are waiting, food and so on. Besides, our system also connects with social media, like facebook and instragam. Let everyone can share information on it.

6. The technical environment of the system

An app may build on Android or iOS.

D. Conceptual Design and Product with Storyboard

我們經由 Task Analysis 所製作出來的 Affinity diagram 來挖掘使用者的痛點 (例如：平日外出吃飯時，多數人都覺得等待是一件很浪費時間的事)，並針對這些痛點進行腦力激盪發想功能。首先我們發現被訪談者在外面吃飯時通常都會攜帶手機，或是在吃飯中間會滑手機，因此我們就決定使用手機的 APP 來當作我們主要的操作介面。

我們發現使用者的痛點主要有三大項，Waiting (等待時間、等待餐點、等待結帳)，Quality (食物的品質、餐廳的服務)，Decision (一群人常常不知道要吃甚麼，決定很久)。針對這幾點，我們進行功能發想，在發想的同時，因為本身組員都具有工程背景 (電機、資工)，所以我們也會考慮的實現的可能性，避免太過於天馬行空。

Scenario 1 & 2

1. Paint point

Waiting (等待時間、等待餐點、等待結帳)

2. Before using our system (Scenario 1)

如果我們有一個程式能在出門前先確認餐廳的地點、餐廳的空位數目，確認完畢之後並能根據通勤的時間做保留位子的動作 (與 Google Map 結合)，並能觀看菜單進行點菜，甚至是事先結帳 (與各方線上支付軟體結合, ex: Apple pay, Paypal)。以下為整個流程的情境示意圖 (storyboard)。確認空位的實作方式，我們有想到在各個椅子上都佈滿 IoT 的壓力感測器或是有些店家會在店內使用監視器觀看不同樓層客人的狀況，我們也可以用電腦視覺的技術對監視器畫面進行分析。

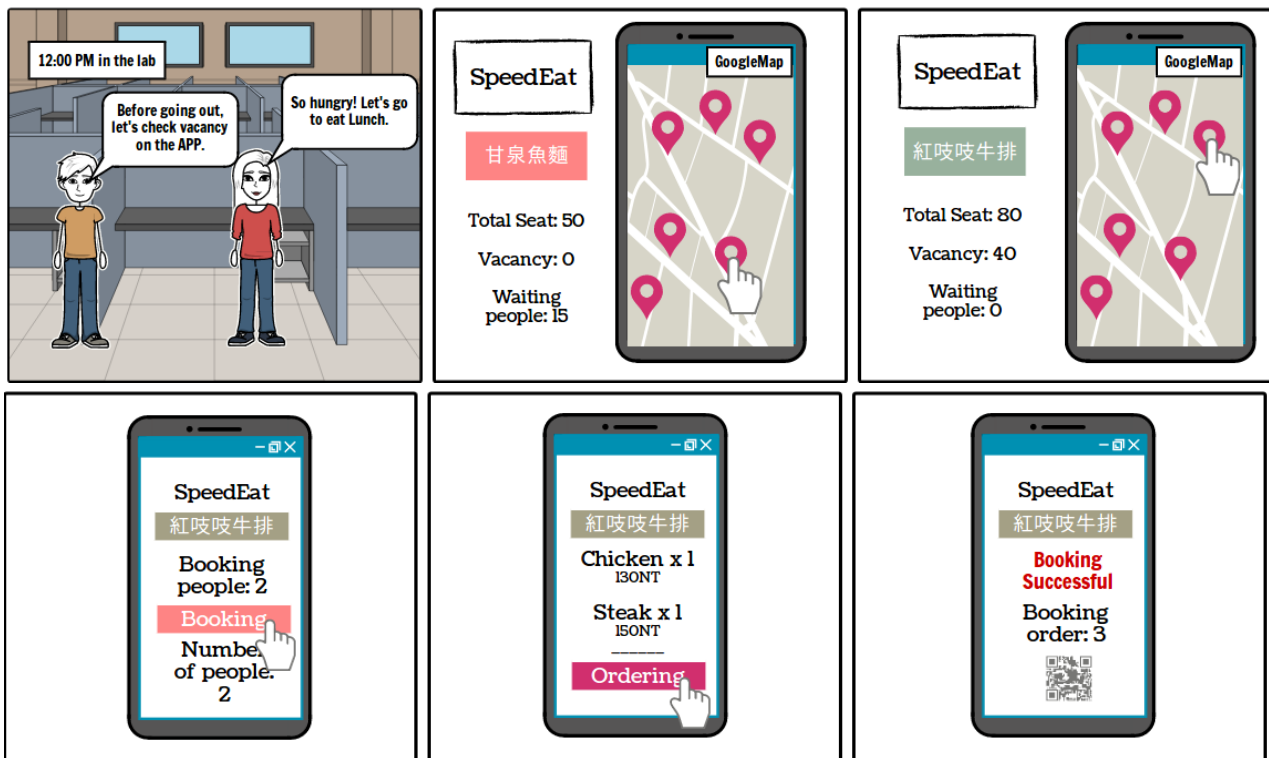
在使用系統前，我們可能會面臨吃飯時間各個店家都大排長龍的情況，最後不得已只後屈就於一些比較不喜歡甚至是不健康的食物 (微波食品)。



3. After using our system (Scenario 1)

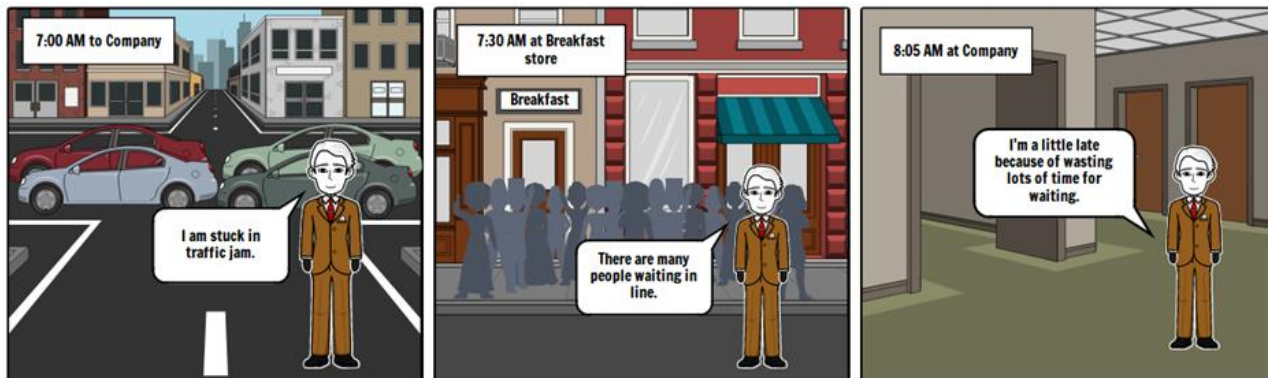
接下來是使用我們系統的情境，在出門前，我們便能先打開 SpeedEat 的 APP 確認各個店家客滿的狀況，像以下的示意圖，我們便能知道還有哪些餐廳目前是比较少人的，甚至我們能根據過去的數據搭配通勤時間來做 10 分鐘或 20 分鐘後人數的預測。選定好餐廳後，我們便能輸入人數來預定位置。我們認為對於大型的餐廳 (價位較高的餐廳) 能直接預定特定位置，但是考慮到小型店家的翻桌率 (價格不高，但等待每組客人要花費太多額外的時間)，我們認為給予預定序號 (booking order) 是比較合理的，就有點像是去診所看病有些人會先預約，如果過號也可以到場會先被優先安排位置。接下來我們就可以觀看菜單，並進行點餐和線上付

費的動作。接下來只要到餐廳之後，就不會浪費太多的時間在找尋空位、等待餐點和等待結帳。



4. Before using our system (Scenario 2)

第二種情境是竹科的上班族，在使用我們的系統前會因為買早餐、塞車而浪費許多時間，甚至導致上班或開會遲到。



5. After using our system (Scenario 2)

但是在使用我們的系統時，可以對於各家早餐店的人潮，及交通狀況進行分析預測，並規畫出最佳的選擇跟路徑，讓上班族節省許多時間，並能準時到達公司。

對於這個功能的實作方式，我們能跟 Google Map 的路線規劃與政府的車流系統結合，達到更精準的預測。



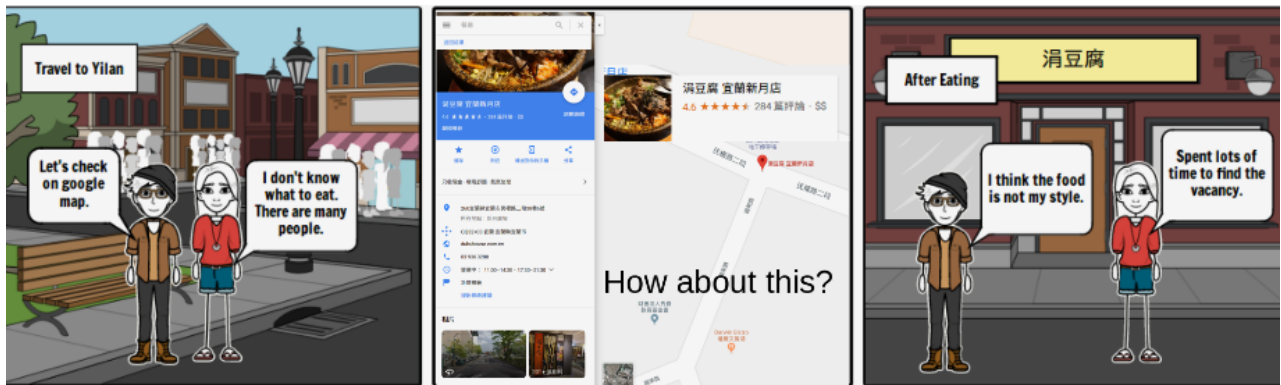
Scenario 3

1. Paint point

Quality (食物的品質、餐廳的服務) and Decision

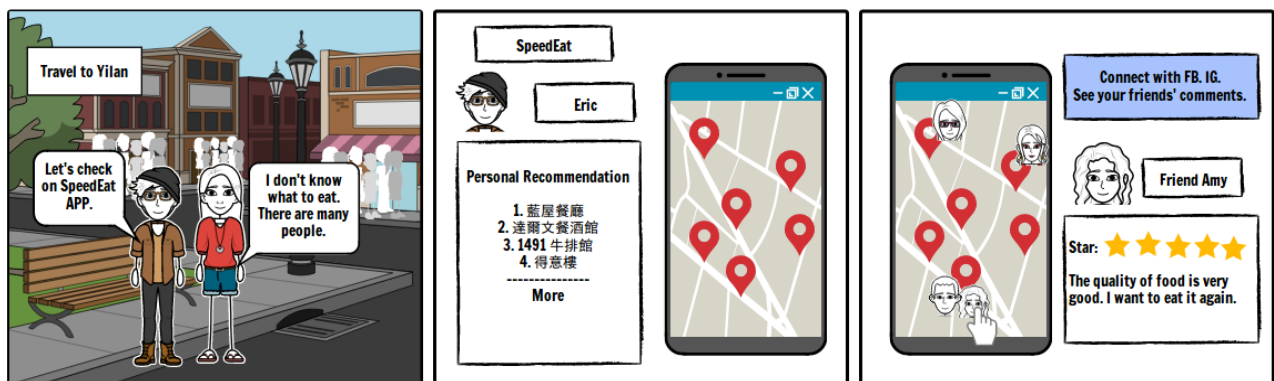
2. Before using our system (Scenario 3)

如果我們在陌生的地方 (旅遊、出差) 常常會不知道可以吃甚麼，有些一開始查好的有名的店又大排長龍，只好在現場查 Google Map，但有時 Google Map 評價不錯的餐廳，吃完後又覺得還好(可能不合胃口或評論太少)。



3. After using our system (Scenario 3)

如果我們能根據過去使用者個人喜好吃的餐廳，來做統計分析，並做最精準的個人推薦，去預測那些餐廳有很高機率會是 user 喜歡的餐廳，甚至可能結合上面的空位機制，同時考慮進等待時間來做推薦。除此之外，現在有許多人喜歡在 facebook 或 ig 分享不同餐廳的食記，但通常這樣的食記比較比較散亂，如果我們可以將之與 google map 整合，你便能看到有哪些朋友或追蹤對象推薦的餐廳。你也會因為了解朋友的口味，而知道誰的喜好與你比較相近，來幫助決策。



Other

老師上課也有提到一些我們可以利用的餐廳類型，像是有些人會想在一個禮拜的幾天中吃的比較清淡，或是哪些是比較適合帶小朋友的餐廳，這些不同的 attribute 都可以幫助使用者更快的做選擇。

<註> 前面我們藉由訪談、persona、使用者分析，去找尋他們共同的痛點且推敲背後的原因，並以 3 種情形描述去 envision system function，並藉由 storytelling 去修改我們的功能。接著是實現的部分，我盡量將我們 envision 的東西通通做出來，並做使用者測試。

E. Software Prototype

The thought process:

After analyzing all the interviews and creating affinity diagram we understood that our basic concept didn't fulfill all the user requirements and needs. Almost half of the users, especially Taiwanese ones (deciding factor was probably ability to speak and read Traditional Chinese), regularly order food via cellphone or using restaurants website and expect it to be delivered to their home.

While our original idea was making users life easier, it didn't bring enough novelty and didn't fasten the process enough. Even if the user was able to find free seats via our app, he/she would still have to go through all the basic activities, like checking the menu, calling the service and ordering. To simplify and hasten this process, we decided to add function that will give the user possibility to not only reserve the seat, but also pre-order the food. We think that this function will benefit not only the user, who will not have to wait for the waiter and food preparation, but also the restaurant owner - the capacity of the restaurant should grow significantly.

We decided to add "I'm in a hurry" and make it our flagship function - the user doesn't have to remember the name of the restaurant or think about the type of the food he/she wants to it. After clicking "I'm in a hurry" button our app will show him/her all the free sits in the restaurants in the chosen area radius and the time to get to all of them.

We decided that we also need to design the panel for the restaurant owner/restaurant service. In a perfect situation, the sits would have the sensors that would be connected to arduino/raspberry pi, which would update the application automatically. The cost of the arduino/raspberry pi and sensors is very low, but, of course, we can't expect the restaurant to buy/have it. In that case we have to design very simple and easy to use interface that will not discourage the service in a situation when the restaurant is crowded.

At this point of the process we started to be worried if our application didn't grow to the point of being messy and hard to navigate, what forced us to create the map of the functions. This action helped us to not only integrate functions with each other in a better way, but also let us know the key paths. We also understood that we have to simplify the interface and the flow in order to encourage first-time users to keep coming back to our app.

Choosing key functions:

1. Client side:

- Finding the restaurant by type of the food/name
- Showing free chairs and the time the user needs to get to the restaurant
- "I'm in a hurry" feature that will show closest restaurants with free seats
- Preview opening hours
- Reserve seat
- Reserve seat and preorder the food

- Order food to home and preview the time needed for it to arrive
- Order using default address
- Order using specified address
- Order food and pay using credit card
- Order food and pay with cash on delivery
- Create list of favorite restaurants
- Check the previous orders
- Share your meals on social media
- Check your friends shares on social media

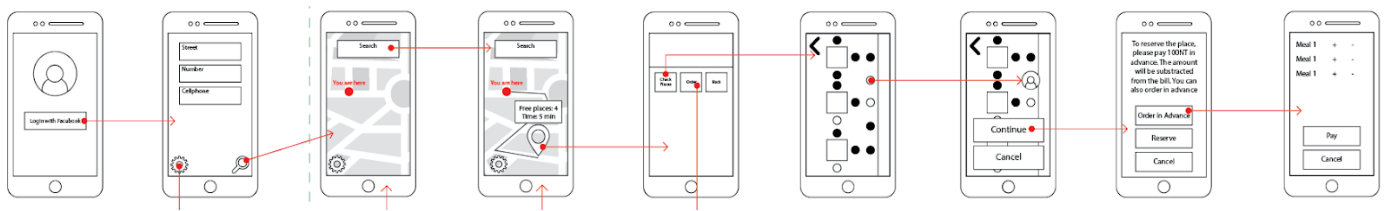
2. Restaurant side:

- Display/edit opening time
- Add/change the arrangement of sits and tables
- Mark sits as taken/free with only one touch/click
- Create/update online menu

Key Paths:

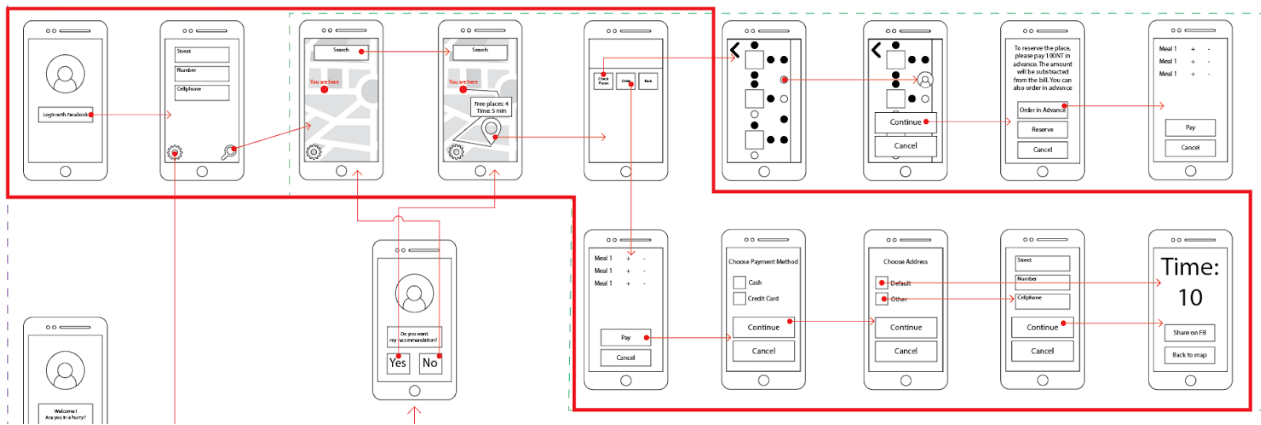
1. Reserving the sit and preordering the food after first login:

The users logs in into application and is asked for the address and cellphone, which will be later used as defaults. Then he/she searches for the restaurant using the type of the food or the name of the restaurant. He/she gets the information about vacancies, and the time needed to get to the place. He/she checks the free places, reserve one, orders in advance (user get prompted about the need of prepaying 100 NT, what should protect the restaurant owner from users that reserve the sits and then never come; user gets informed that this amount will be subtracted from the bill) and then chooses the meal according to the menu



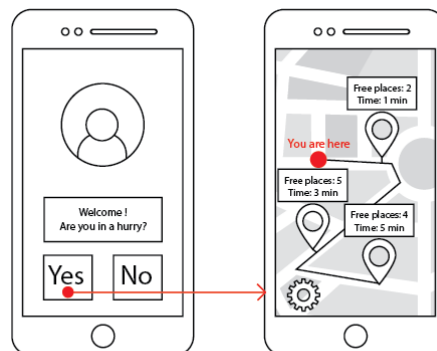
2. Ordering the food using default address after first login:

The users logs in into application and is asked for the address and cellphone, which will be later used as defaults. Then he/she searches for the restaurant using the type of the food or the name of the restaurant. He/she touches the marker, gets to the restaurant screen, touches "Order" button, choses meal, payment method, address (can choose default or add other. After paying, the user can see the time the order needs to be executed, which is based on the Google Traffic API.

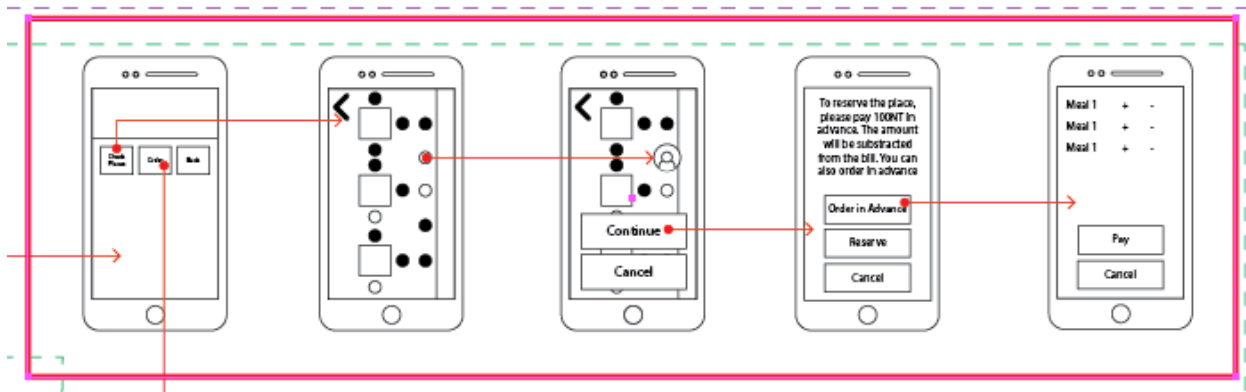


3. Regular user login , “I’m in a hurry”, reserve the seat and preorder food:

In the moment of login the user is asked if he/she is in a hurry. If yes, he/she is directed to the screen that shows all the free places in the restaurants in chosen radius from the place he/she currently is.

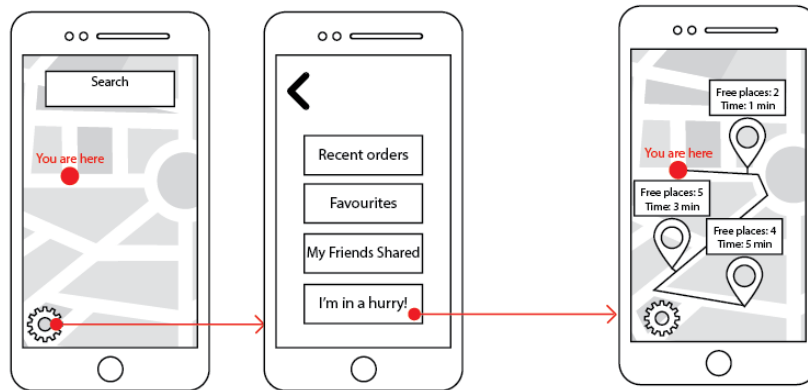


After that the reservation of the seat and pre ordering the food goes exactly like after first login.



4. Regular user logged, while searching for the restaurant decided that he/she is in a hurry:

While searching for the restaurant the user decides he/she is in a hurry. He/she clicks the option button and from there is directed to the screen showing free places in the restaurants in the chosen area radius.

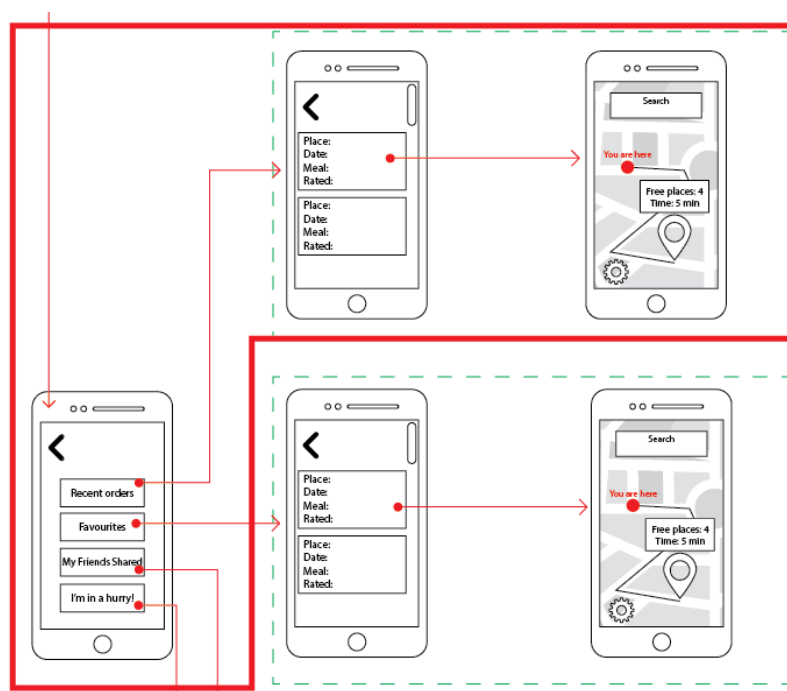


5. Regular user login, not in a hurry, reserve and preorder:

In the moment of login the user is asked if he/she is in a hurry. If no, he/she is directed to the regular screen with map and search button.

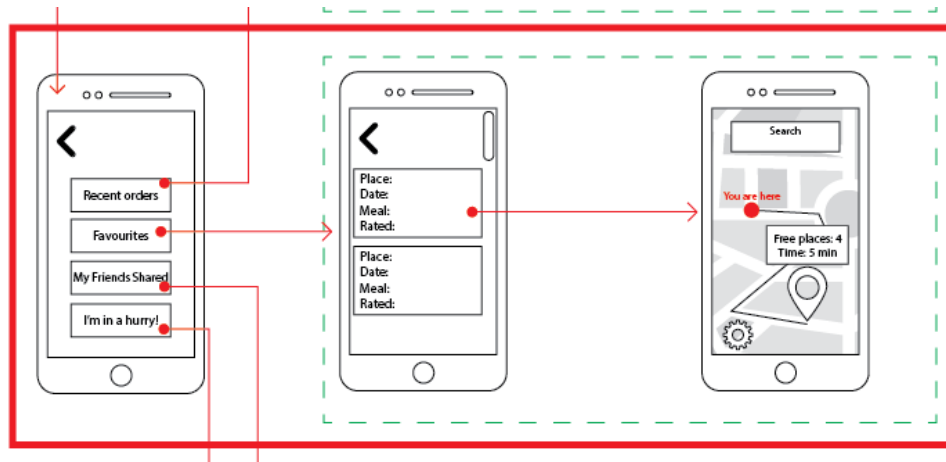
6. User wants to check his recent orders:

User uses the option button, the touches/clicks on the recent orders and is directed to the screen that shows them ordered by the time. If he clicks one of them, he is directed to the screen showing how far he/she is from the place and the free places. After that he/she can go back, do the regular search or reserve/order from this restaurant.



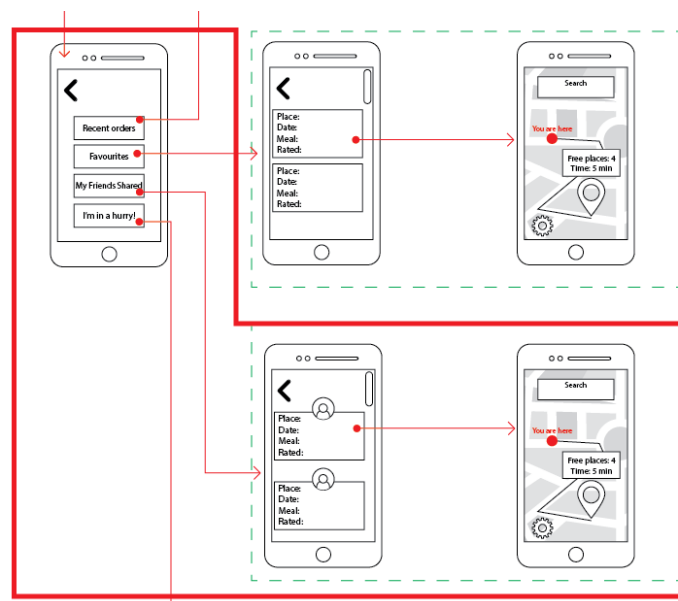
7. User wants to check how far he/she is from his favorite places:

User clicks/touches the options button, then touches the favorites button. After that he/she is directed to the list that shows his favorite places. If he/she clicks on one of them he/she is directed to the map that shows the distances and number of free places.



8. User wants to check the places his/her friends went to and shared the locations:

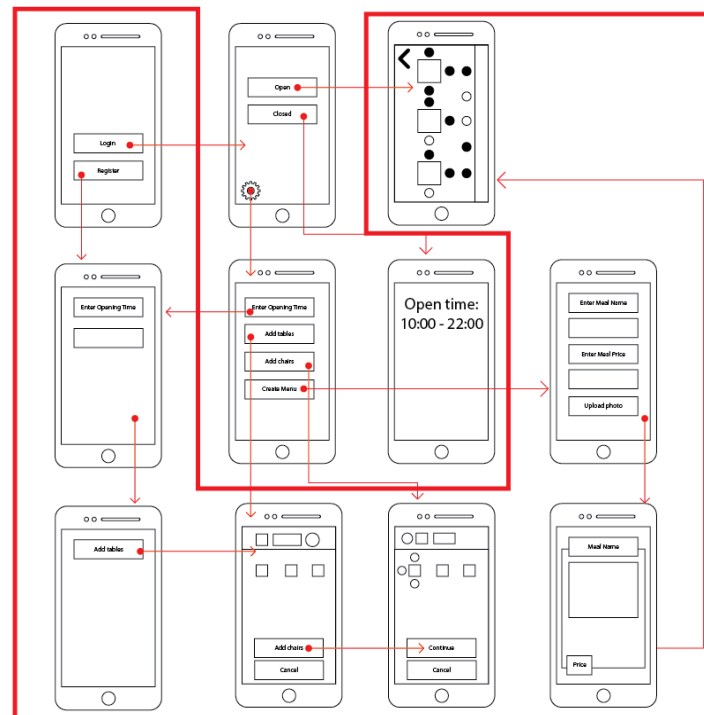
User clicks/touches the options button, then my friends shared button. After that he/she is directed to the list with his/her friends shares and he/she can clicks on one of the places what will direct him/her to the map showing the distance and free sits.



- Restaurant registration:

This process should be as simple and intuitive as possible to not discourage the restaurant from using our application. After clicking register button the user will be prompt to enter restaurant opening time. In case the restaurant is closed and the user tries to order something, he will see the screen with opening hours. After this, the restaurant owner will be prompt to add tables, chairs and menu. After finishing the process, the restaurant owner will be directed to the screen showing vacancies. In our design, the user have to click/touch the chair to mark it as "taken". If the user

clicks/touches the “taken” sit, it becomes marked free. The user can add/change tables, sits and menu at any time given via options menu.



F. User Test

Motivation

Before starting to implement the prototype we decided to make a real life user testing. We chose to use the paper prototype not only because we were very curious after watching videos in class, but also we think that this is a great way of testing key functions and paths, because it tests not only the user behavior, but also our own knowledge. We thought that if user will be able to navigate without a problem and if we won't spend too much time thinking which board we should give next, it means that the design is good.

Test design

1. Model preparation:

Basing on the function map we created, we designed the boards which should be printed on the paper and then cut out by us. Besides the screens, we printed numbers, user symbols, sits symbols, map, keyboard and markers.



We also prepared the stand for the cell phone using duct tape, pipe and cardboard, because we thought that if one of us will stand over the users head with the cellphone, the user might have problems with focus.

2. Test design:

Test includes four different tasks and usage of our key-functions on the users side:

- Login with facebook, add address and cellphone, search restaurant using type of a food “Taiwanese”, reserve the seat
- While already having the account, search for the “Japanese” food, reserve the seat, pre-order the food
- Testing “I’m in a hurry” function up to the place reservation
- While already having the account, order the food to users place

3. Conclusion:

After the test we briefly interviewed the user and asked for her input.

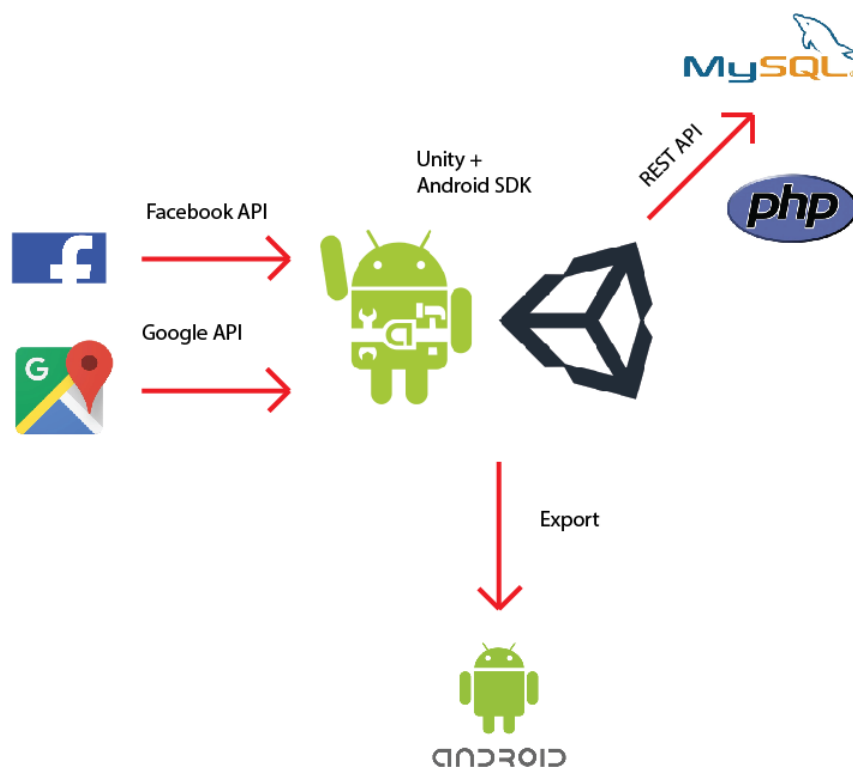
- The user was a bit confused, because she never participated in application test (the usage of cardboard was a bit confusing at first especially) what made the test more viable
- The user thought that, after first test, it was easy to navigate
- We didn't have the possibility of adding default address and credit card. User suggested that this is "a must have" so we went back and redesigned the functions map. The necessity of entering the same data multiple times is very irritating
- It was very easy for us to navigate through our application and give the user the right boards.

G. Prototype preparation

Tools

After the user testing we started implementing our prototype. We couldn't find any prototyping software that was sophisticated enough for our needs. We decided to at least try to check if our idea is possible to implement by us. We chose Unity, because it is possible there to use all the API's that we would need: Google Traffic, Google Maps, Facebook API. We also managed to configure Android SDK (our target platform) what makes the export possible. Additionally we created MySQL database and wanted to connect it with Unity using REST API written in PHP.

Architecture



Implementation

Unfortunately we overestimated the time we had for testing prototype creation. We were able to implement the touch functions, rotation and moving the map and choosing restaurant. In the separate project we also managed to create facebook login. We were also able to export our project to Android.

H. Appendix (Interview results)

Context inquiry

We not only ask the question but also let the interviewer tell their experience of eating outside.

First part interview

Hi, I am ... and I wanted to ask you a few questions about your experiences with going out for a meal. We are trying to understand people's needs for the Interaction Design class final project. None of the data collected during the interview will be used outside of the class. Please remember that you can terminate the interview at any time. If any of the questions seems to personal for you or you simply don't want to answer, please don't hesitate to tell me. Is this ok with you? And please imagine you are going to eat outside.

Interview1:

1. What are the troubles of eating at outside?

Don't know what to eat. Time cost of waiting for meals. Time cost of waiting in line. Food is not good. The dishes are not clean. The food is not greasy. Don't want to eat with strangers. The CP value of food is not high.

2. What is the most trouble thing to eat?

Don't know what to eat.

3. How long does it take to eat at outside?

30 min

4. How do you decide what to eat?

Weather, last meal

5. How long does it take to commute for eating?

15 min (weekdays)

6. Will you want to wait long time for food?

Yes, 5 min

7. Do you eat yourself?

No, with friend

8. Is waiting to be bothering thing?

Yes

9. If you are in the new and unfamiliar environment, how do you decide what to eat?

Number of waiting people, restaurant rating (google map, star, etc.)

10. Will you make appointment in advance?

Just for big meal. (For breakfast store, I will book in advance because I have to wait)

12. If you are hurry, how do you decide what to eat?

The faster food.

13. Parking space of restaurant is also an important problem.

14. Do you decide what to eat according to how long need to wait?

Yes

15. How do you select restaurant in order to not be late for next event?

According to my experience.

16. Do you have specially preferred foods?

Yes, hot pot, braised pork rice

17. Do you have bad experience of eating outside?

It take long time to wait for meal. And I will go to ask clerk.

18. How do you book in advance?

1. Phone Call 2. Very few experience about online reservations. Because I don't know whether it's successful booking through online system.

19. Have you used Uber EAT or Similar app?

Yes, less experience.

20. Have you called the delivery?

Yes, Less

21. The procedure to decide what to eat.

No one knows what to eat -> Someone suggested restaurant -> Occasionally rebutted

22. What do you do during eating?

Use smart phone.

23. How do you pay?

Easy card ->cash ->credit card

24. Wait in line to checkout -> It's an annoying thing.

25. Are Checkout first, or post checkout the bothering things?

Not a troublesome thing

26. Do you have experience of being late because of eating?

Yes, waiting for meal.

27. Do you have the experience of wasting a lot of time waiting to turn the table?

Yes, in department store.

Interview2:

1. What are the troubles of eating?

Don't know what to eat. Too much choice. Too many people. Need to wait for long time.

2. What is the most trouble thing to eat?

Don't know what to eat.

3. Who do you usually eat with?

Friends

4. How long does it take to eat?

1hr

5. How do you decide what to eat?

Number of waiting people, Weather, Price, Drink

6. How long does it take to commute for eating ?

15 min

7. Will you want to wait long time for food?

No

8. Is waiting to be bothering thing?

Yes, unless for delicious food.

9. If you are in the new and unfamiliar environment, how do you decide what to eat?

Price, others' comments

10. Will you make appointment in advance?

Just for a big meal

11. If you are hurry, how do you decide what to eat?

The faster the better, microwave food

13. Do you decide what to eat according to how long need to wait? Yes

14. How do you select restaurant in order to not be late for next event?

According to my experience

15. Do you have specially preferred foods?

Black tea

16. Do you have bad experience of eating outside?

Eat something bad or very expensive. Take long time to wait.

17. How do you book restaurant in advance?

Phone Call, No internet reservation

18. Have you used UberEAT or similar app?

NO

19. Have you called the delivery?

Yes, phone call

20. The procedure to decide what to eat.

suggested restaurant-> Occasionally rebutted

21. What do you do during eating?

Use smart phone

22. How do you pay?

Cash

23. Wait in line to check out -> OK, usually acceptable

24. Are Checkout first, or post checkout the bothering things?

No, but payment first is better.

25. Do you have experience of being late because of eating?

Yes, because of too many people or waiting too long for food

26. Do you have the experience of wasting a lot of time waiting to turn the table?

Yes

Interview3:

1. What are the troubles of eating?

The food is not good to eat. Service is not good. Food is too little. Too few choices because of most of food is not good to eat.

2. What is the most trouble thing to eat?

The food is not good to eat.

3. Who do you usually eat with?

Friends or alone

4. How long does it take to eat?

20 min

5. How do you decide what to eat?

Service, Price, good to eat or not

6. How long does it take to commute for eating?

20 min

7. Will you want to wait long time for food?

No. Only if food is great.

8. Is waiting to be bothering thing?

Yes, unless for delicious food.

9. If you are in the new and unfamiliar environment, how do you decide what to eat?

Something looks special or just choose which food he always eats before.

10. Will you make appointment in advance?

Just for a big meal

11. If you are hurry, how do you decide what to eat?

The faster the better

13. Do you decide what to eat according to how long need to wait?

Yes

14. How do you select restaurant in order to not be late for next event?

According to my experience

15. Do you have specially preferred foods?

Burritos

16. Do you have bad experience of eating outside?

Eat something bad or service is bad

17. How do you book restaurant in advance?

Phone Call, No internet reservation

18. Have you used UberEAT or similar app?

NO

19. Have you called the delivery?

Seldom

20. The procedure to decide what to eat.

Personally: distance, appetite. Group: Always hart to decide

21. What do you do during eating?

Use smart phone or Observing others

22. How do you pay?

Cash

23. Wait in line to check out taking too many time.

-> OK, usually acceptable

24. Are Checkout first, or post checkout the bothering things?

No.

25. Do you have experience of being late because of eating?

Yes, because of waiting too long for food

26. Do you have the experience of wasting a lot of time waiting to vacancy?

Yes

Interviewer4:

1. What are the troubles of eating?

Don't know what to eat. Need to wait for long time because of waiting for vacancy .

2. What is the most trouble thing to eat?

Don't know what to eat.

3. Who do you usually eat with?

Friends

4. How long does it take to eat?

40 min

5. How do you decide what to eat?

Number of waiting people, mood, food is good to eat or not, price

6. How long does it take to commute for eating?

20 min

7. Will you want to wait long time for food?

Seldom 5min

8. Is waiting to be bothering thing?

Yes, especially when I am hurry.

9. If you are in the new and unfamiliar environment, how do you decide what to eat?

Others' comments on google

10. Will you make appointment in advance?

Just for a big meal

11. If you are hurry, how do you decide what to eat?

The faster the better

13. Do you decide what to eat according to how long need to wait? Yes

14. How do you select restaurant in order to not be late for next event?

According to my experience, comment on google

15. Do you have specially preferred foods?

Anything good to eat is ok.

16. Do you have bad experience of eating outside?

Eat something bad or very expensive. Take long time to wait for vacancy or meal.

17. How do you book restaurant in advance?

Phone Call, No internet reservation

18. Have you used UberEAT or similar app?

NO

19. Have you called the delivery?

No

20. The procedure to decide what to eat.

suggested restaurant-> Occasionally rebutted

21. What do you do during eating?

Use smart phone

22. How do you pay?

Cash

23. Wait in line to check out -> Yes

24. Are Checkout first, or post checkout the bothering things?

No.

25. Do you have experience of being late because of eating?

Yes, because of too many people or waiting too long for food

26. Do you have the experience of wasting a lot of time waiting to vacancy?

Yes

Second part interview

Hi, I am ... and I wanted to ask you a few questions about your experiences with going out for a meal. We are trying to understand people's needs for the Interaction Design class final project. None of the data collected during the interview will be used outside of the class. Please remember that you can terminate the interview at any time. If any of the questions seems to personal for you or you simply don't want to answer, please don't hesitate to tell me. Is this ok with you?

1) How often do you eat outside?

2) What makes you decide if you want to eat in a certain place?

3) Do you have favorite places?

4) Are they often crowded?

5) How often you find yourself waiting in line?

6) Follow up question – if yes, then how long do you usually wait?

7) Can the empty table, so you don't have to wait, be an important factor in choosing the place to eat?

8) Let's imagine the situation that you are very hungry so you are going to your favorite place to eat, but you can see that it is full and you will have to wait. Would you wait or look for the place with an empty sit?

- 9) Let's imagine the situation that you have the application you can use to check if there are any free places in your favorite restaurant. Would you use it and check before leaving home?
- 10) Would you prefer the restaurant that lets you check free places by using the application over the one you have to go and check yourself?

Interviewer5

- 1) I eat outside every single day. (Follow up- how many times a day?) 1 or 2 per day.
- 2) If I like the food or not. Nice menu with tempting pictures of food outside
- 3) Yes. For example Vietnamese restaurant next to RT Mart
- 4) Yes, especially during weekends
- 5) I usually don't wait in line. If the place is very crowded I'm choosing other place.
- 6) I can wait maximum 10 minutes
- 7) Yes. Actually it is, quite often.
- 8) Most of the times I'd look for a different place
- 9) Yes, it sounds very useful
- 10) If the quality of the food would be good. The quality of food is still the most important, but I'd love to have application like this.

Interviewer6

- 1) 4 times per week, during working days mostly, because I don't have the time to cook during the week
- 2) Most important things for me are: price, speed of food preparation and taste
- 3) I've been living in Taiwan for a very long time already so yes, I have some.
- 4) Yes. Sometimes I have to wait for a very long time, but they serve me very quickly when I get the place finally.
- 5) Rarely, but definitely it happens, usually if I go to the lunch/dinner punctually like 12 pm and 7 pm.
- 6) - 7) I usually take food to go, but if I want to eat inside than for sure I don't want to wait, especially during the winter.
- 8.) Sometimes I wait but usually I don't want to wait. I don't have the time for that because I have to go back to work so I'm in a hurry.
- 9) Yes, I would definitely use application like this.
- 10) I'm not sure, I'd have to have the application first to see if there is any difference.

Interviewer7

- 1) I eat outside very often, almost every day. Sometimes even 2-3 times a day.
- 2 The proximity of the place from my workplace and the availability of good food. Of course, the

price is also very important because I'm a student.

3) Yes, I have favorite places.

4) Yes, They are often crowded. A lot of people go there. Especially foreigners.

5) A lot of times. Especially during lunch time.

6) It can be anywhere between 10 minutes to half an hour.

7) Yes, That would be a very important factor to decide where to eat.

8.) Depends on the day, if I can afford some time then I would wait. Otherwise, I would just go to the place with an empty seat. It depends on the availability of time.

9) Yes . Definitely. This sounds like a very good idea.

10) Yeah. I am OK with that. I would definitely use it.